

Pre-Interview Task

Instructions

Write/draw/etc a recommended two pages. All candidates are given the same task. You will not receive extra credit for fast or long submissions. Responsible use of AI or other tools is welcome --- focus on answering the questions to the best extent. You are not marked on writing style/grammar. You are marked more on your ability to summarise a lot of knowledge and ideas into what is most relevant. You are marked on the criteria at the end of the document.

Problem Statement

Existing workflows have grown significantly and evolved independently. They now contain inconsistent configuration behaviour, tooling choices, logging standards, retry logic, and security assumptions. You are tasked with proposing an improved architecture of the Python and C++ SDK workflows with the view of rolling out a solution across other Hiero repositories. Answer these questions, providing concrete recommendations specific to the problem at Hiero. Any research should be relevant and thoughtfully applied.

- 1) Outline the principles of a well architected software system and consider how these could be applied in your solution [4] [K=2, R=2]
- 2) Draw a diagram of your proposed final system architecture at a high level [4] [R=2,A=2]
- 3) Identify the key architectural boundaries and justify why they exist. [4] [R=2, E=2]
- 4) Justify your choices of any tooling the app will need to support these components [6] [R=2, A=2, E=2]
- 5) Briefly explain how this solution can be built iteratively [2] [A=2]
- 6) A user creates a malicious pull request in the C++ or Python SDK that uses this app with the view of exfiltrating information. Given your proposed architecture, explain any risks and which safeguards can be applied against such threats. [10] [K=4, R=3, A=3]
- 7) The C++ and Python have made different configuration choices. Explain which configuration options you will standardise, and which configuration options you allow as user inputs, subject to avoiding an overly high maintenance tax. [6] [K=2, R=2, A=2]
- 8) Provide an actionable plan on how you can leverage the Hiero community to better achieve the app's goals. [10] [K=3, R=3, A=4]
- 9) Assess how many weeks, if any, should be allocated to system architecture, or if development should commence immediately. [4] [R=2, E=2]

Total = 50

Marking criteria:

- [K] Knowledge: accuracy, depth and breadth of understanding.
- [R] Relevance: the ability to identify best-fit information and prioritise
- [A] Analysis: the ability to break down a problem into logical steps
- [E] Evaluation: weighing up relative costs and benefits to form justified, logical conclusions